Staff Augmentation Scope of Work

Tech Services Section

September 2026

This Staff Augmentation Scope of Work directly supports Line of Effort 1: Website, Database, and Technology in the Tech Services Work Plan for FY2027, whose desired end state is that technology resources used by the Division are up-to-date, new applications can be rolled out quickly, and technical debt is minimized. Achieving that end state requires dedicated outside expertise to assess DWRi's current infrastructure and chart a modernization path, which is precisely the discovery, assessment, and roadmapping work scoped in this engagement.

This engagement also directly advances the commitments and findings of three related efforts. It operationalizes the technology modernization priorities set out in the Utah Division of Water Rights FY 2027 Action Plan. It responds to the data-integrity, program-practice, and planning deficiencies identified in the Utah Office of the Legislative Auditor General's 2024 performance audit. It builds on and extends the USU Technology Modernization Roadmap, applying dedicated staff augmentation capacity to carry out the roadmap's sequenced task groups for modernizing DWRi's core software, database, and data-governance systems.

Phase 1 — Discovery & Assessment (Months 1–4)

- Review prior reports and project references (see below) with DWRi and DTS staff and USU; confirm scope and priorities, and reflect on the technical debt identified in the USU Hydroinformatics and Technology Gap Analysis.
- Review the “User-Centric Design Requirements for the Water Measurement Data Dashboard” document and the relevant Figma mockups, and propose development approaches for the dashboard, including the software architecture and technology stack.
- Evaluate the current MapServer's architecture, dependencies, and integration options with up to date ArcGIS Maps SDK for JavaScript.
- Evaluate and assess the degree that the HydroServer architecture, GCP environment, and API designs could be used as a pattern or to otherwise inform other division systems.
- Assess DWRi's web infrastructure, including its legacy Classic ASP stack, technical debt, security, and third-party dependencies; evaluate modern web-platform options against ADA compliance, Utah Design System adoption, mobile responsiveness, cross-browser compatibility, maintainability, and support needs.

- Assess Classic ASP, Python 2.7, and FORTRAN codebases individually — inventory scope, usage, dependencies, and retirement priority for each, rather than treating "legacy stack" as a single undifferentiated assessment.
- Assess the Division's approximately 47 interconnected SQL Server databases — evaluate the current lack of defined primary/foreign keys, use of triggers and monolithic journal tables, and the feasibility and sequencing of transitioning to an API-first data-access model where only the API connects directly to the data.
- Assess the "Other Software Development" catalog — the undocumented FORTRAN, VB.NET, and Python batch scripts and scheduled tasks running on the Batch Server — and propose an approach to inventory and document their existence, ownership, and function.
- Assess personnel and staffing structure and skills gaps, including the current divide between developers working exclusively in the modern Flask/React environment and longer-tenured developers who have not yet migrated from legacy environments.
- Assess current authentication and authorization and the feasibility of transitioning to UtahID, including a proposed roles and permissions model.
- Review Internal staff and external user experience feedback via [interviews / surveys / usage analytics] to identify pain points and priority fixes.
- Evaluate potential integration pathways for Google Gemini (LLM), strictly within the bounds of the State of Utah AI policy, including data governance and appropriate-use constraints.

Phase 2 — Recommendations & Roadmap (Month 3–6)

- Document current-state architecture and findings from Phase 1.
- Propose a target architecture for DWRi's software tools and services, with particular focus on upcoming custom HydroServer front-end tools.
- Develop and propose a modernization approach for MapServer.
- Propose a modernized web development pipeline — version control, CI/CD, and test/production parity — currently absent for Classic ASP, FORTRAN, and Python 2.7 code.
- Propose a personnel/staffing plan alongside the technical roadmap, recommending which workstreams are best owned by Division staff (including DTS programmers) versus outside contractors.
- Include a recommendation for website/subdomain consolidation or restructuring (e.g., an informational CMS site, an api subdomain, and an apps subdomain).
- Include recommended sequencing for new application builds — online application forms, searchable scanned-document system — alongside legacy-system migration priorities.

- Propose a phased modernization implementation plan that addresses prioritized technical debt and gaps identified through the USU studies and Phase 1 assessment, including timeline, sequencing, dependencies, relative priority/effort, user-testing approach, legacy-system decommissioning strategy, and ongoing support/ownership considerations.

Deliverables

#	Deliverable	Acceptance Criteria
1	A document and diagrams describing current-state infrastructure assessment, with flowcharts/schematics of DWRi's technical architecture	Documents major systems, dependencies, integrations, risks, and findings from Phase 1
2	A document and diagrams describing proposed target architecture for DWRi software/services, including custom HydroServer front-end tools	Addresses security, ADA, UtahID, GCP/HydroServer integration, LLM constraints, and documented key user requirements.
3	A document describing MapServer modernization proposal	Includes options, trade-offs, and a recommended approach
4	A document describing Gantt Chart timeline for a phased modernization implementation roadmap.	Addresses prioritized technical debt and gaps, sequencing, dependencies, relative effort, user testing, and legacy-system decommissioning
5	Personnel/staffing recommendation document	Identifies staffing gaps and skills needs; recommends staff vs. contractor ownership per workstream
6	Database modernization proposal	Covers key/relational structure, API-first access transition, and data-platform consolidation options
7	New application requirements document	Scopes online well log submittal, online application forms, and searchable scanned-document system

Grant him access to the following

- GitHub https://github.com/utahdwri and our repos at https://github.com/utahdnr
- HydroServer https://hydroserver.waterrights.utah.gov/
- ClickUp https://clickup.com/
- State gmail account + VPN access + Guest access to SQL Server
- Google Cloud Platform (GCP) https://console.cloud.google.com/ and DWRi projects

Project References

The consultant shall review and leverage the following resources throughout the engagement to inform recommendations, avoid duplication of effort, and ensure alignment with the Division's modernization objectives

1. Horsburgh, J. S., Slaugh, D., & Lippold, K. (2024). Utah Division of Water Rights: Technology Modernization Roadmap. Utah State University, Logan, UT. HydroShare. https://doi.org/10.4211/hs.6bb192887d474555a75f1df86e9a9f66
2. HydroServer web. https://hydroserver.waterrights.utah.gov/ and HydroServer Documentation (API and TypeScript clients). https://hydroserver2.github.io/hydroserver/
3. Existing HydroServer APIs and technical documentation, including OpenAPI specifications and Python client documentation. https://hydroserver2.github.io/hydroserver/
4. Utah Office of the Legislative Auditor General (2024). A Performance Audit of Ensuring Data Integrity, Program Best Practices, and Comprehensive Water Planning. https://waterrights.utah.gov/strategic-plan/DWRi-Strategic-Plan-2026.pdf
5. Utah Division of Water Rights Action Plan 2026–2027. https://waterrights.utah.gov/strategic-plan/DWRi-Strategic-Plan-2026.pdf
6. User-Centric Design Requirements for the Water Measurement Data Dashboard https://docs.google.com/document/d/1f2-MGF65XH14v-5As8qBViDeEphH5l0gCJjcZJmGZHI/edit?usp=sharing
7. Utah Division of Water Rights GitHub Organization. https://github.com/utahdwri
8. UtahID. About UtahID. https://my.utah.gov/u?id=about

9. Utah Design System: https://designsystem.utah.gov/ and https://github.com/utahdts/utah-design-system
10. Water Rights MapServer https://maps.waterrights.utah.gov/EsriMap/map.asp?layersToAdd=distribution
11. Utah Geospatial Resource Center (UGRC). https://gis.utah.gov/
12. The UGRC Discover Geospatial Data Catalog. https://discover.agrc.utah.gov/
13. ArcGIS Maps SDK for JavaScript https://developers.arcgis.com/javascript/latest/